Red Wine Quality: Reproducible Data Workflow

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Overview

This report documents a reproducible exploratory workflow for the UCI red Vinho Verde wine-quality dataset. The workflow loads the source CSV, applies transparent cleaning functions, computes descriptive summaries, and creates three figures that describe patterns rather than causal effects. The notebook, source data, requirements file, and this report are distributed together so another analyst can rerun the analysis.

Dataset Description

The raw red-wine file contains 1,599 rows and 12 columns: eleven physicochemical measurements and a quality score from 0 to 10. After removing 240 exact duplicate records, the analysis uses 1,359 rows. The UCI documentation notes that the data describe Portuguese Vinho Verde samples and do not include variables such as producer, grape type, brand, price, or full sensory context (Cortez et al., 2009). These omissions limit the scope of any conclusions.

Workflow Description and Key Decisions

The workflow follows five stages: ingestion, cleaning, exploratory analysis, visualization, and interpretation. Column names are standardized for consistent reuse in code, and exact duplicates are removed to avoid overweighting repeated records in counts and averages. The analysis then calculates descriptive statistics, groups alcohol content by quality, and examines correlations with quality. Keeping the data, executable notebook, package versions, and version-control history together supports a reproducible workflow, consistent with the use of documented code and computational environments described by Danchev (2022) and Stoudt et al. (2021).

Results and Interpretation

Figure 1. Distribution of Alcohol Content

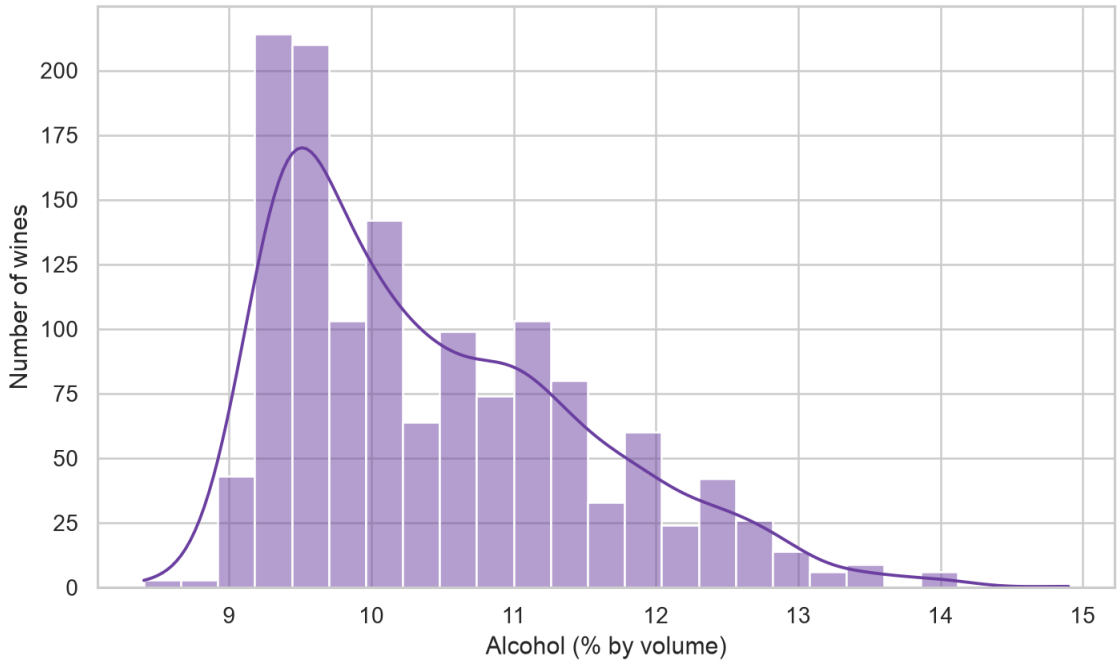


Figure 1 shows that alcohol content is concentrated around the center of its observed range. The shape is not evidence that alcohol causes quality, but it motivates comparing alcohol across quality groups.

Figure 2. Alcohol Content by Wine Quality Score

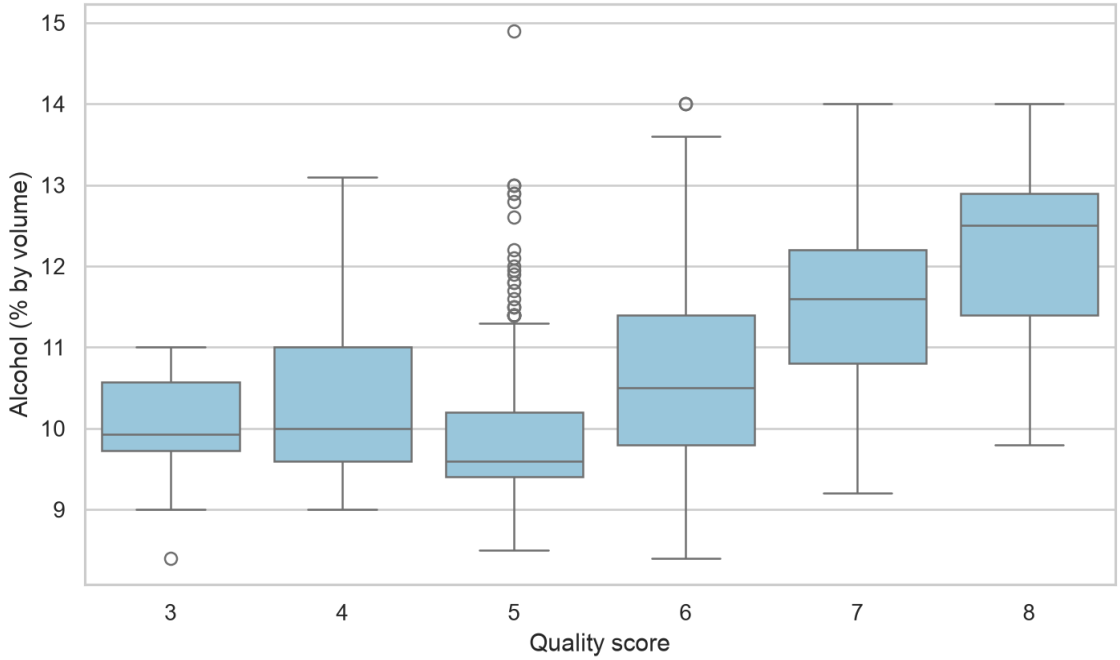


Figure 2 shows that the central alcohol level tends to rise with quality score, although boxes overlap substantially. Individual chemistry variables alone cannot fully explain the sensory score.

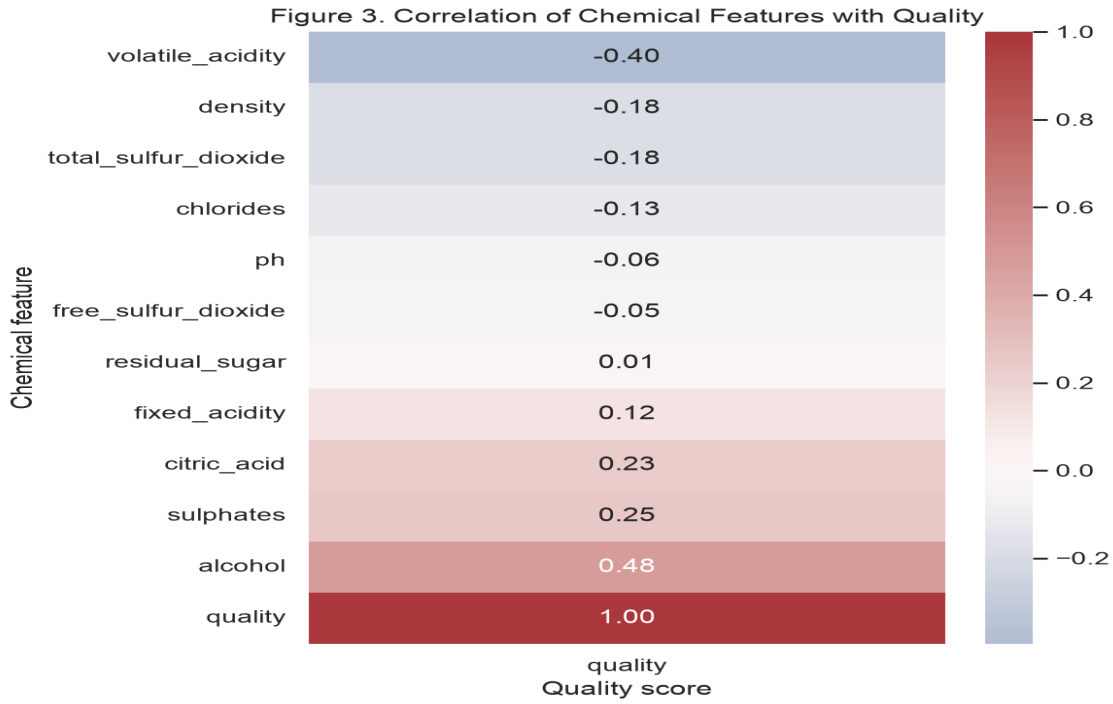


Figure 3 summarizes linear associations. Alcohol is positively associated with quality, while volatile acidity is among the strongest negative associations. Correlations describe co-movement, not causal effects.

Responsible Practice: Bias and Data Quality

Cleaning can itself introduce bias if records are removed without a defensible rule or if missing values are imputed in a way that changes group relationships. Here, only exact duplicates are removed and the source has no missing cells. However, the dataset's limited context and imbalanced quality categories mean that it should not be treated as a representative sample of all wines. Future work should document collection procedures, inspect subgroup coverage, and retain an audit trail of every transformation.

Reproducibility

A reviewer can recreate the analysis by installing packages from requirements.txt and running data_workflow.ipynb from top to bottom in the repository root. The project uses a dedicated development branch and multiple commits so changes to cleaning, EDA, and documentation remain traceable. This separation of exploratory work and polished output follows the staged workflow perspective described by Stoudt et al. (2021).

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, 47*(4), 547-553. https://doi.org/10.1016/j.dss.2009.05.016

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education, 5*(56), 156. https://doi.org/10.21105/jose.00156

Stoudt, S., Vasudevan, V., & Higgins, R. A. (2021). Principles for data analysis workflows. *PLOS Computational Biology, 17*(3), e1008770. https://doi.org/10.1371/journal.pcbi.1008770